

## MOHYLIV-PODIL'S'KYI, Ukraine

WMO: 336630

Lat: 48.4523N

Lon: 27.7797E

Elev: 256

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |            |            | 99.6%                         |            |            | 99%        |            |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>1.3 | (c)<br>8.8 | (d)<br>-2.9                   | (e)<br>4.7 | (f)<br>2.2 | (g)<br>3.0 | (h)<br>6.5 | (i)<br>10.1 | (j)<br>12.5           | (k)<br>33.8 | (l)<br>10.0 | (m)<br>32.7 | (n)<br>1.4               | (o)<br>290 | (p)<br>0.445 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>22.6            | (c)<br>91.3     | (d)<br>70.0 | (e)<br>88.0 | (f)<br>68.8 | (g)<br>85.2 | (h)<br>67.9 | (i)<br>72.8         | (j)<br>86.0 | (k)<br>71.0 | (l)<br>83.8 | (m)<br>69.5 | (n)<br>81.3 | (o)<br>4.1           | (p)<br>140 |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.5                            | 105.9 | 78.4 | 66.9 | 99.9 | 76.1 | 65.5 | 95.1 | 74.6 | 36.5          | 85.6 | 35.0 | 84.0 | 33.7 | 81.5 | 83.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| 10.8              | 9.3  | 7.8 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
|                   |      |     | -4.0                       | 96.0 | 7.2                | 3.7 | -9.1                                               | 98.6 | -13.3      | 100.8 | -17.3      | 102.9 | -22.5      | 105.5 |
|                   |      |     | -4.4                       | 75.8 | 7.1                | 2.3 | -9.5                                               | 77.4 | -13.7      | 78.8  | -17.6      | 80.1  | -22.8      | 81.7  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.0   | 27.8  | 30.7  | 39.4  | 51.3  | 61.1  | 67.3  | 70.9  | 69.5  | 60.4  | 49.6  | 40.5  | 30.7  |  |  |
|                                                                        | DBStd       | 17.01  | 9.40  | 9.74  | 8.09  | 7.13  | 6.68  | 5.77  | 5.26  | 5.62  | 6.58  | 7.52  | 8.12  | 9.45  |  |  |
|                                                                        | HDD50       | 2639   | 687   | 539   | 338   | 68    | 5     | 0     | 0     | 0     | 4     | 99    | 300   | 599   |  |  |
|                                                                        | HDD65       | 5994   | 1152  | 959   | 794   | 413   | 157   | 43    | 11    | 21    | 167   | 478   | 735   | 1064  |  |  |
|                                                                        | CDD50       | 2649   | 0     | 0     | 8     | 106   | 348   | 518   | 647   | 604   | 315   | 88    | 15    | 0     |  |  |
|                                                                        | CDD65       | 532    | 0     | 0     | 0     | 2     | 36    | 111   | 193   | 160   | 28    | 2     | 0     | 0     |  |  |
|                                                                        | CDH74       | 6016   | 0     | 0     | 0     | 53    | 537   | 1262  | 2045  | 1745  | 352   | 22    | 0     | 0     |  |  |
|                                                                        | CDH80       | 2155   | 0     | 0     | 0     | 10    | 140   | 407   | 811   | 698   | 86    | 3     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.0    | 3.1   | 3.3   | 3.7   | 3.4   | 3.0   | 2.8   | 2.5   | 2.2   | 2.5   | 2.6   | 3.2   | 3.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 24.1   | 1.4   | 1.3   | 1.1   | 1.8   | 2.9   | 3.5   | 3.7   | 2.2   | 1.7   | 1.2   | 1.6   | 1.6   |  |  |
|                                                                        | PrecMax     | 35.9   | 6.2   | 3.8   | 2.7   | 4.0   | 8.1   | 8.4   | 8.9   | 4.3   | 5.6   | 4.0   | 5.9   | 3.0   |  |  |
|                                                                        | PrecMin     | 14.6   | 0.1   | 0.2   | 0.0   | 0.4   | 0.8   | 0.6   | 0.7   | 0.4   | 0.0   | 0.2   | 0.2   | 0.2   |  |  |
|                                                                        | PrecStd     | 5.2    | 1.1   | 1.0   | 0.7   | 0.9   | 1.7   | 2.0   | 2.0   | 1.0   | 1.6   | 0.9   | 1.2   | 0.8   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 49.6  | 55.3  | 67.2  | 79.6  | 88.0  | 91.3  | 96.6  | 96.0  | 87.2  | 76.7  | 65.3  | 53.6  |  |  |
|                                                                        |             | MCWB   | 43.7  | 45.9  | 52.1  | 60.2  | 65.6  | 70.0  | 71.2  | 72.1  | 67.0  | 62.8  | 56.2  | 48.4  |  |  |
|                                                                        | 2%          | DB     | 44.3  | 49.7  | 61.2  | 74.0  | 84.0  | 87.6  | 91.5  | 91.2  | 82.3  | 70.2  | 59.1  | 49.0  |  |  |
|                                                                        |             | MCWB   | 40.8  | 43.7  | 49.2  | 57.1  | 64.8  | 68.7  | 70.2  | 70.0  | 64.1  | 59.3  | 53.3  | 45.7  |  |  |
|                                                                        | 5%          | DB     | 41.7  | 45.9  | 56.5  | 70.0  | 79.9  | 84.8  | 88.0  | 87.4  | 78.1  | 65.5  | 55.6  | 45.0  |  |  |
|                                                                        |             | MCWB   | 38.9  | 41.7  | 46.5  | 55.3  | 63.1  | 67.7  | 69.5  | 68.7  | 62.9  | 56.9  | 51.4  | 42.5  |  |  |
|                                                                        | 10%         | DB     | 39.2  | 42.4  | 52.3  | 66.0  | 76.1  | 81.8  | 84.6  | 83.6  | 73.8  | 61.2  | 52.4  | 41.7  |  |  |
|                                                                        |             | MCWB   | 36.9  | 38.7  | 44.5  | 53.1  | 61.4  | 66.6  | 68.7  | 67.8  | 61.1  | 54.8  | 48.9  | 39.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 44.1  | 47.4  | 53.0  | 61.6  | 69.0  | 73.7  | 75.0  | 75.4  | 69.0  | 64.6  | 57.1  | 49.8  |  |  |
|                                                                        |             | MCDB   | 48.2  | 53.4  | 64.3  | 76.9  | 83.0  | 85.7  | 87.5  | 89.5  | 83.5  | 75.1  | 62.8  | 52.6  |  |  |
|                                                                        | 2%          | WB     | 41.5  | 44.4  | 50.2  | 58.7  | 66.6  | 71.1  | 72.9  | 72.6  | 66.2  | 60.6  | 54.1  | 45.7  |  |  |
|                                                                        |             | MCDB   | 43.8  | 48.6  | 59.4  | 71.1  | 79.5  | 83.1  | 86.3  | 86.4  | 78.3  | 68.4  | 58.2  | 48.6  |  |  |
|                                                                        | 5%          | WB     | 39.1  | 41.8  | 47.7  | 56.3  | 64.6  | 69.5  | 71.3  | 70.6  | 64.0  | 58.0  | 51.8  | 42.8  |  |  |
|                                                                        |             | MCDB   | 41.3  | 45.7  | 54.7  | 67.3  | 76.4  | 81.1  | 84.0  | 83.4  | 74.5  | 64.2  | 55.3  | 45.1  |  |  |
|                                                                        | 10%         | WB     | 37.0  | 39.1  | 45.4  | 54.2  | 62.5  | 67.8  | 69.8  | 68.7  | 62.3  | 55.4  | 49.0  | 39.5  |  |  |
|                                                                        |             | MCDB   | 39.0  | 42.0  | 51.4  | 63.8  | 73.7  | 78.5  | 81.5  | 80.0  | 71.7  | 60.3  | 52.1  | 41.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.9   | 12.3  | 16.6  | 21.0  | 23.6  | 22.4  | 22.6  | 23.6  | 20.9  | 17.4  | 11.0  | 9.8   |  |  |
|                                                                        |             | MCDBR  | 12.1  | 17.7  | 24.9  | 28.9  | 30.0  | 28.1  | 29.3  | 30.4  | 28.8  | 24.7  | 17.0  | 12.9  |  |  |
|                                                                        | 5% WB       | MCWBR  | 9.1   | 12.4  | 14.5  | 14.1  | 13.5  | 11.7  | 11.0  | 11.5  | 12.9  | 14.0  | 10.9  | 9.8   |  |  |
|                                                                        |             | MCWBR  | 11.3  | 16.4  | 22.5  | 25.5  | 26.2  | 24.0  | 24.3  | 26.2  | 24.6  | 22.3  | 15.0  | 12.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.324 | 0.356 | 0.395 | 0.455 | 0.422 | 0.420 | 0.452 | 0.456 | 0.412 | 0.378 | 0.355 | 0.322 |  |  |
|                                                                        | taud        |        | 2.223 | 2.170 | 2.174 | 2.063 | 2.234 | 2.266 | 2.188 | 2.171 | 2.249 | 2.319 | 2.304 | 2.288 |  |  |
|                                                                        | Ebn at Noon |        | 226   | 245   | 256   | 251   | 267   | 268   | 256   | 249   | 248   | 237   | 214   | 212   |  |  |
|                                                                        | Edh at Noon |        | 25    | 33    | 39    | 48    | 43    | 41    | 44    | 43    | 36    | 29    | 23    | 21    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 325   | 574   | 963   | 1411  | 1800  | 1907  | 1858  | 1656  | 1150  | 717   | 318   | 239   |  |  |
|                                                                        | RadStd      |        | 52    | 94    | 107   | 139   | 149   | 147   | 117   | 111   | 160   | 97    | 33    | 42    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
|                         | +1.19 | N/A     | N/A    | N/A     | N/A   | N/A   | -232        | -342  | +202  | +99   |
| Regional (20 neighbors) | +1.21 | N/A     | N/A    | +1.42   | N/A   | N/A   | N/A         | -353  | +214  | +95   |

Nomenclature: See separate page